Outlier Remove Practical

There is dataset of winequality

Need to find out outlier and remove it from the dataset

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
df = pd.read_csv('winequality.csv')
df.head()
```

```
df.describe()
```

- We will use the residual sugar column to detect and remove the outliers

we will plot the data

```
sns.distplot(df['residual sugar'])
```

- There is a outlier as the plot is completely right skewed

Next we will use boxplot to see the outliers clearly

```
sns.boxplot(df['residual sugar'])
```

- The box represents the interquartile range (IQR), with the line inside representing the median. The whiskers extend to the minimum and maximum values within 1.5 times the IQR from the first and third quartiles. Any points outside of the whiskers are considered potential outliers

Methods to remove Outliers

Z-Score Method

- The z-score method is a statistical technique used to detect outliers by measuring how many standard deviations a data point is away from the mean. A z-score tells you how relatively far a data point is from the mean in terms of standard deviations

First we will get the upper and lower limits

```
upper_limit = df['residual sugar'].mean() + 3*df['residual sugar'].std()
lower_limit = df['residual sugar'].mean() - 3*df['residual sugar'].std()
print('upper limit:', upper_limit)
print('lower limit:', lower_limit)
```

- These limits define a range beyond which data points are considered outliers based on the z-score method

find outliers using the limits

```
df.loc[(df['residual sugar'] > upper_limit) | (df['residual sugar'] <
lower_limit)]
```

- The resulting DataFrame outliers will contain only the rows where outliers are present in the '**residual sugar**' column

Next we will trim the outliers. Trimming is a data transformation technique where outliers are removed or "trimmed" from the dataset

```
new_df = df.loc[(df['residual sugar'] <= upper_limit) & (df['residual sugar'] >=
lower_limit)]
```

```
new_df.head()
```

```
new_df.reset_index(inplace=True)
```

plot the data after trimming outliers

```
sns.boxplot(new_df['residual sugar'])
```

Next we will perform capping. Capping, also known as Winsorization, is a technique used to handle outliers by setting a threshold and capping or truncating extreme values to a specified percentile. Capping involves replacing outlier values with less extreme values, thus reducing the impact of outliers on the dataset without entirely removing them

```
df2 = new_df.copy()
```

```
df2.loc[(df2['residual sugar']>=upper_limit),'residual sugar']=upper_limit
df2.loc[(df2['residual sugar']<=lower_limit),'residual sugar']=lower_limit
```

- By performing capping the outlier values in the '**residual sugar**' column are replaced with the specified upper or lower limit values, effectively bringing them within the desired range

Next let us plot the data after performing capping

```
sns.boxplot(df2['residual sugar'])
len(df2)
```

Inter Quartile Range Method

- The **Interquartile Range** (IQR) method is another statistical technique used to detect and handle outliers in a dataset. The IQR represents the range between the first quartile (Q1) and the third quartile (Q3) of a dataset

First calculate the first quartile (Q1), third quartile (Q3), and interquartile range (IQR) of the 'residual sugar' column in the DataFrame df

```
q1 = df['residual sugar'].quantile(0.25)
q3 = df['residual sugar'].quantile(0.75)
iqr = q3-q1

q1, q3, iqr
```

Next let us calculate the upper and lower limit using the Interquartile Range (IQR) method

```
upper_limit = q3 + (1.5 * iqr)
lower_limit = q1 - (1.5 * iqr)
lower_limit, upper_limit
```

Next let us plot the data

```
sns.boxplot(df['residual sugar'])
```

Next we will find the outliers using upper and lower limit calculated earlier

```
df.loc[(df['residual sugar'] > upper_limit) | (df['residual sugar'] <
lower_limit)]
```

Next let us perform trimming of the outliers

```
new_df = df.loc[(df['residual sugar'] <= upper_limit) & (df['residual sugar'] >=
lower_limit)]
print('before removing outliers:', len(df))
print('after removing outliers:',len(new_df))
print('outliers:', len(df)-len(new_df))
```

```
new_df.reset_index(inplace=True)
```

Next let us plot the data after trimming outliers

```
sns.boxplot(new_df['residual sugar'])
```

Next let us perform capping of the outliers

```
new_df = df.copy()
new_df.loc[(new_df['residual sugar']>upper_limit), 'residual sugar'] = upper_limit
new_df.loc[(new_df['residual sugar']<lower_limit), 'residual sugar'] = lower_limit
```

Next let us plot the data after performing capping

```
sns.boxplot(new_df['residual sugar'])
```